Master's Thesis

Practical Regularized Quasi-Newton Methods with Inexact Function Values (不正確な関数値下における実用的な正則化準ニュートン法)

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Abstract

Many practical optimization problems involve objective function values that are corrupted by unavoidable numerical errors. In smooth nonconvex optimization, quasi-Newton methods combined with line search are widely used due to their efficiency and scalability. These methods implicitly assume accurate function evaluations and thus may fail to converge in noisy settings. Developing fast and robust quasi-Newton methods for such scenarios is therefore crucial.

To address this issue, we propose a noise-tolerant regularized quasi-Newton method equipped with a relaxed Armijo-type line search, designed to remain stable under inaccurate function evaluations. By combining a regularization parameter update rule inspired by Objective-Function-Free Optimization and the AdaGrad-Norm method, we establish a global convergence rate of $\mathscr{O}\left(1/\varepsilon^2\right)$ for reaching a first-order stationary point under the assumed error model. We performed extensive experiments on the CUTEst benchmark collection with artificially noisy objective function evaluations, as well as with low-precision floating-point arithmetic (64-, 32-, and 16-bit). The results demonstrate that the proposed method is substantially more robust than several existing methods, while maintaining competitive practical convergence speed and computational cost.

Keywords Regularized Quasi-Newton Method, Numerical Stability, Numerical Error, AdaGrad-Norm, Objective-Function-Free Optimization

Contents

Chapter 1

Introduction

1.1 Introduction

We consider the unconstrained optimization problem

$$\underset{x\in\mathbb{R}^n}{\text{minimize}} \quad f(x), \tag{1.1}$$

where $f\colon \mathbb{R}^n \to \mathbb{R}$ is a continuously differentiable nonconvex function, and $x \in \mathbb{R}^n$ is the optimization variable. We focus on settings where objective function evaluations are contaminated by bounded and non-diminishing numerical noise [46]. Specifically, only inexact function values $\overline{f}(x)$ are available for optimization. Such inaccuracies are unavoidable in many realistic scenarios, including finite-precision arithmetic, simulation-based evaluations, and stochastic approximations [9, 36, 45]. When function values are contaminated by numerical noise, standard optimization algorithms may lose their theoretical guarantees, behave erratically, or terminate prematurely with unreliable solutions.

In the absence of noise and with accurate function evaluations, quasi-Newton methods, in particular the Broyden–Fletcher–Goldfarb–Shanno (BFGS) method and its Limited-memory variant (L-BFGS), are among the most efficient and widely used algorithms to solve the problem (1.1) [31, 38]. Their success is largely attributed to fast local convergence and scalability. Standard implementations [39, 49] rely on line search procedures that enforce the Wolfe conditions to guarantee sufficient descent and the positive definiteness of the Hessian approximation.

In the presence of numerical noise, these line search conditions may become unreliable. Differences in function values or directional derivatives can be dominated by numerical errors, leading to unstable step sizes, ill-conditioned Hessian approximations, or abnormal termination [40, 45]. From a computational perspective, improving the robustness of quasi-Newton methods in noisy environments is therefore crucial. In particular, it is desirable to design algorithms that (i) retain computational efficiency when function evaluations are accurate, (ii) remain stable when function values are unreliable, and (iii) guarantee convergence to first-order stationary points under mild assumptions.

In this work, we propose a noise-tolerant regularized quasi-Newton method that addresses these challenges. Our approach combines Armijo-type line search with quadratic regularization [27, 48] and Objective-Function-Free Optimization (OFFO) inspired strategy [22, 23]. When a sufficient decrease in function value is observed, the method behaves similarly to standard quasi-Newton methods with line search, preserving its practical efficiency. When numerical noise dominates, we introduce regularization on the approximate Hessian to ensure numerical stability, which yields convergence properties similar to those of OFFO methods, particularly AdaGrad-Norm [51]. The resulting algorithm is particularly well-suited for low-precision or noisy environments.

From a theoretical standpoint, we establish that the proposed method achieves the standard global convergence rate of $\mathscr{O}\left(1/\varepsilon^2\right)$ for smooth nonconvex optimization problems, even when objective function values are corrupted by noise. Extensive numerical experiments on the CUTEst benchmark collection, across various floating-point precisions and artificial noise settings, demonstrate that the proposed method is significantly more stable than standard methods with line search in the presence of significant numerical noise, while maintaining practical convergence speed.

1.2 Organization of This Thesis

The remainder of this thesis is organized as follows.

Chapter 2 reviews the foundations of quasi-Newton methods. In section 2.1, we briefly explain what quasi-Newton methods are. In section 2.2, we explain the modified secant condition, which is used in our numerical experiments. These materials are presented in this chapter, as they are standard and well documented in the literature.

Chapter 3 presents the proposed regularized quasi-Newton method along with its theoretical analysis. In section 3.1, we review relevant background and introduce the problem setting. In section 3.2, we present the proposed algorithm. In section 3.3, we establish a global convergence property of the proposed algorithm. In section 3.4, we discuss practical choices of algorithmic variables and parameters. In section 3.5, we report numerical results on the CUTEst benchmark collection. Finally, in section 3.6, we conclude the thesis with a summary and future research directions.

Our implementation is publicly available on GitHub[*1].

[*1] https://github.com/HirokiHamaguchi/qnlab

Chapter 2

Foundations of Quasi-Newton Methods

In this chapter, we review the foundations of quasi-Newton methods.

2.1 Quasi-Newton Methods

In this section, we discuss quasi-Newton methods. Quasi-Newton methods are based on Newton's method but aim to reduce its primary drawback, namely the high computational cost of evaluating the Hessian matrix. Instead of using the true Hessian matrix $\nabla^2 f(x_k)$, an approximation matrix B_k is employed, thereby achieving fast convergence while keeping the computational cost low.

Figure 2.1 shows a comparison of Newton's method, quasi-Newton methods, and gradient descent. Gradient descent updates the iterate in the direction opposite to the gradient. Although its per-iteration computational cost is the lowest, its convergence is slow. Newton's method requires the fewest iterations, but it has the highest computational cost per step. Quasi-Newton methods lie between these two extremes and strike a balance between them. In particular, when the methods are compared in terms of total computation time rather than the number of iterations, quasi-Newton methods often exhibit the best performance.

Quasi-Newton methods with line search generate a sequence $\{x_k\}_k$ defined by

$$x_{k+1} = x_k - \alpha_k B_k^{-1} \nabla f(x_k) = x_k - \alpha_k H_k \nabla f(x_k).$$

Here, $\alpha_k > 0$ is the step size determined by line search, B_k is an approximation of the Hessian matrix $\nabla^2 f(x_k)$ at the point x_k, and $H_k \coloneqq B_k^{-1}$ denotes its inverse. This update rule closely resembles that of Newton's method, which is why it is called a quasi-Newton method.

The core of quasi-Newton methods lies in how to update B_k at each iteration so that it approaches the true Hessian matrix. Figure 2.2 illustrates this concept. First, around the current point x_k, a quadratic approximation model of the objective function f is constructed using B_k. Next, this quadratic model is minimized to obtain the next point x_{k+1}. After obtaining x_{k+1}, the approximation matrix B_k is updated to B_{k+1} using the gradient information at x_k and x_{k+1}. This procedure is repeated until convergence, which constitutes the quasi-Newton methods.

2.1.1 Secant Condition

Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be of class C^2. Suppose that a symmetric matrix B_k is given as an approximation of the Hessian matrix $\nabla^2 f(x_k)$. We consider updating the approximate Hessian to B_{k+1} for the next point x_{k+1}. Although there are infinitely many possible candidates for such a matrix, it is natural to impose symmetry on B_{k+1}, since the true Hessian matrix is symmetric. For the current point x_k and the next point x_{k+1}, we define the step and the gradient difference as follows:

$$s_k \coloneqq x_{k+1} - x_k, \qquad y_k \coloneqq \nabla f(x_{k+1}) - \nabla f(x_k).$$

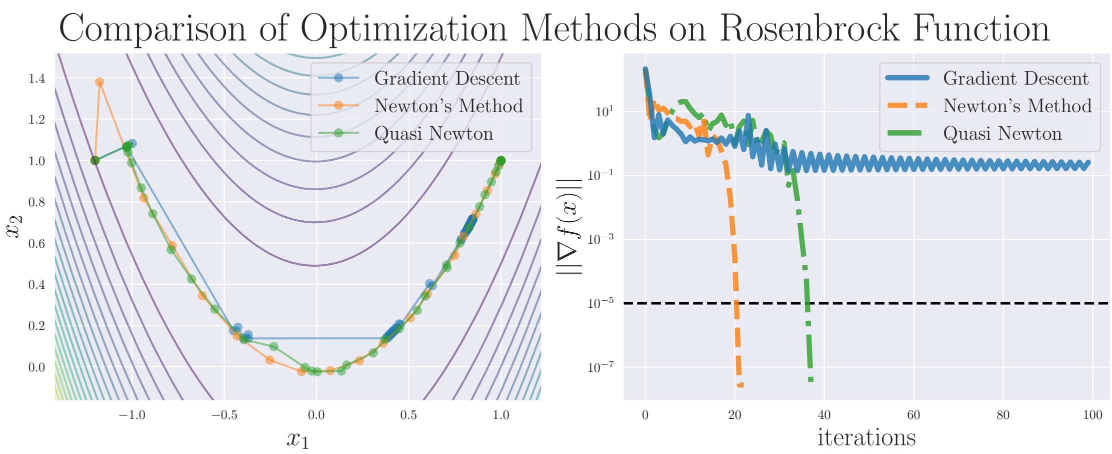


Fig. 2.1: Comparison of Newton's method, quasi-Newton methods, and gradient descent.

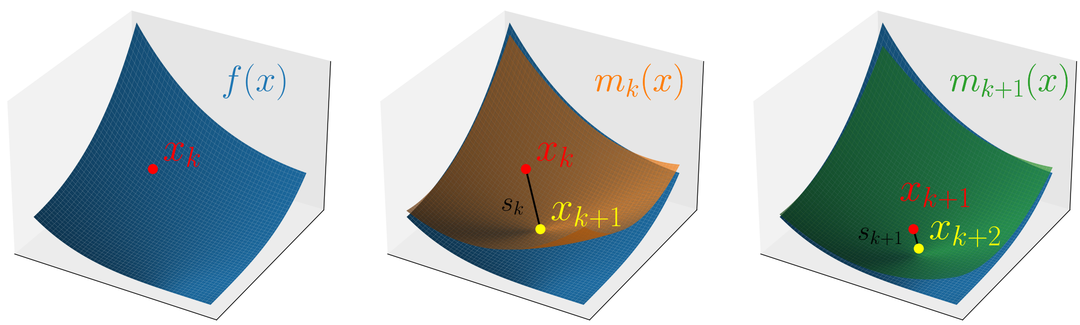


Fig. 2.2: Conceptual illustration of quasi-Newton methods. (1) Given the objective function f (blue surface) and the current point x_k (red point). (2) Move to the minimizer x_{k+1} (yellow cross) of the quadratic model $m_k(x)$ (orange surface) constructed using the current approximate Hessian. (3) Update the quadratic model (green surface) and repeat this procedure.

In the following, we generally assume that $s_k \neq 0$ and $y_k^\top s_k \neq 0$. Note that the s in s_k stands for "step." Since the new approximation is constructed around x_{k+1}, we approximate $\nabla f(x_k)$ using the Hessian matrix $\nabla^2 f(x_{k+1})$ and its approximation B_{k+1} as follows:

$$\begin{aligned}\nabla f(x_k) &\approx \nabla f(x_{k+1}) + \nabla^2 f(x_{k+1})(x_k - x_{k+1}) \\ &\approx \nabla f(x_{k+1}) + B_{k+1}(x_k - x_{k+1}).\end{aligned}$$

Requiring the above approximation to hold as an equality yields

$$B_{k+1}(x_{k+1} - x_k) = \nabla f(x_{k+1}) - \nabla f(x_k).$$

Equivalently, this can be rewritten as

$$B_{k+1} s_k = y_k.$$

This relation is called the secant condition or the quasi-Newton equation.

More precisely, the secant condition can also be justified as follows. Consider a quadratic approximation model around x_{k+1}:

$$m_{k+1}(x) = f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B_{k+1}(x - x_{k+1}).$$

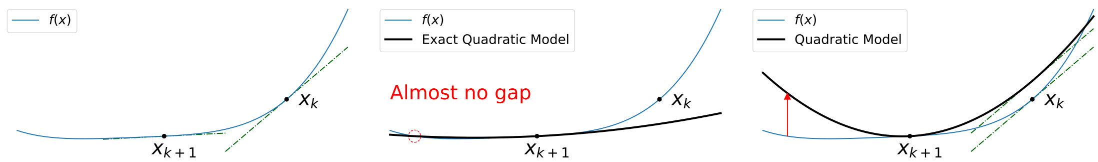


Fig. 2.3: Limitations of the standard secant condition. (a) Function values and gradients at x_k and x_{k+1} are already known. (b) The ideal quadratic model constructed from the exact Hessian fits well around the new point x_{k+1}. (c) The standard secant condition may fail to capture curvature adequately.

By construction, regardless of B_{k+1}, this model satisfies

$$\begin{cases} m_{k+1}(x_{k+1}) = f(x_{k+1}), \\ \nabla m_{k+1}(x_{k+1}) = \nabla f(x_{k+1}). \end{cases}$$

In addition, the secant condition ensures that the model satisfies

$$\begin{aligned} \nabla m_{k+1}(x_k) &= \nabla f(x_{k+1}) - B_{k+1}(x_{k+1} - x_k) && \text{(definition of the model)} \\ &= \nabla f(x_{k+1}) - (\nabla f(x_{k+1}) - \nabla f(x_k)) && \text{(secant condition)} \\ &= \nabla f(x_k). \end{aligned}$$

This implies that the secant condition ensures that the quadratic model $m_{k+1}(x)$ accurately reflects the known information at both x_{k+1} and x_k, except for the previous function value $f(x_k)$.

2.2 Modified Secant Condition

This section explains the modified secant condition, a modification of the standard secant condition. The modified secant condition incorporates function value information in addition to gradient information, leading to more accurate Hessian approximations.

2.2.1 Derivation of the Modified Secant Condition

Although the secant condition is widely used in many quasi-Newton methods, it can fail to capture the curvature of the objective function accurately because it only utilizes gradient information. We illustrate this limitation in Figure 2.3. In Figure 2.3, we have two points x_k and x_{k+1} where function values and gradients are already known. We see that the ideal quadratic model constructed from the exact Hessian fits well around the new point x_{k+1}, leading to a better convergence behavior. However, the standard secant condition uses only gradient information and ignores function values. Consequently, it may significantly misestimate curvature, leading to a poor approximation of the true objective function.

We can overcome this limitation by utilizing the function values. The basic idea is illustrated in Figure 2.4. Even when the gradients are the same at two points x_k and x_{k+1}, natural interpolation of function values differs depending on the function values $f(x_k)$ and $f(x_{k+1})$. This observation motivates the modified secant condition, which incorporates function value information to improve Hessian approximation.

In the following, we present two well-known modified secant conditions.

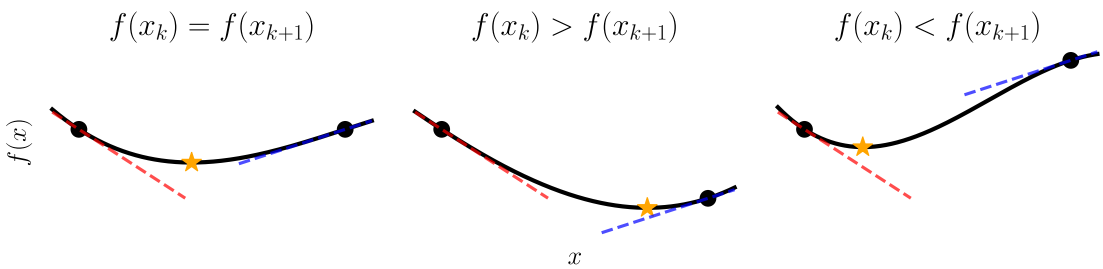


Fig. 2.4: Basic idea of the modified secant condition. Even when the gradients are the same at two points x_k and x_{k+1}, natural interpolation of function values differs depending on the function values $f(x_k)$ and $f(x_{k+1})$.

Function-Value-Matching Modified Secant Condition

The first modification enforces the quadratic model to match the function value at the previous point [1, 52, 56]. Let us consider an alternate model with a different approximate Hessian B^{F}_{k+1}:

$$m^{\mathrm{F}}_{k+1}(x) := f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B^{\mathrm{F}}_{k+1}(x - x_{k+1}).$$

We require this model to satisfy

$$m^{\mathrm{F}}_{k+1}(x_k) = f(x_k),$$

which enforces that the function value at the previous point is correctly modeled.

Let us derive the corresponding modified secant condition for B^{F}_{k+1}. Substituting the definition of m^{F}_{k+1} and using $s_k = x_{k+1} - x_k$, the condition becomes

$$\begin{aligned} f(x_k) &= f(x_{k+1}) + \nabla f(x_{k+1})^\top (x_k - x_{k+1}) + \frac{1}{2}(x_k - x_{k+1})^\top B^{\mathrm{F}}_{k+1}(x_k - x_{k+1}) \\ &= f(x_{k+1}) - \nabla f(x_{k+1})^\top s_k + \frac{1}{2} s_k^\top B^{\mathrm{F}}_{k+1} s_k. \end{aligned}$$

Now, instead of the standard secant condition, we assume a modified form with an additional term involving the identity matrix:

$$\left(B^{\mathrm{F}}_{k+1} + \sigma^{\mathrm{F}}_k I\right) s_k = y_k.$$

Here, $\sigma^{\mathrm{F}}_k \in \mathbb{R}$ is a scalar to be determined as follows. Substituting this equation yields

$$f(x_k) = f(x_{k+1}) - \nabla f(x_{k+1})^\top s_k + \frac{1}{2} s_k^\top y_k - \frac{\sigma^{\mathrm{F}}_k}{2} \|s_k\|^2.$$

Solving for σ^{F}_k and using $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, we obtain

$$\begin{aligned} \sigma^{\mathrm{F}}_k &= \frac{2(f(x_{k+1}) - f(x_k)) - (2\nabla f(x_{k+1}) - y_k)^\top s_k}{\|s_k\|^2} \\ &= \frac{2(f(x_{k+1}) - f(x_k)) - (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2}. \end{aligned}$$

Therefore, the modified secant condition becomes

$$B^{\mathrm{F}}_{k+1} s_k = \hat{y}^{\mathrm{F}}_k := y_k + \frac{2(f(x_k) - f(x_{k+1})) + (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2} s_k.$$

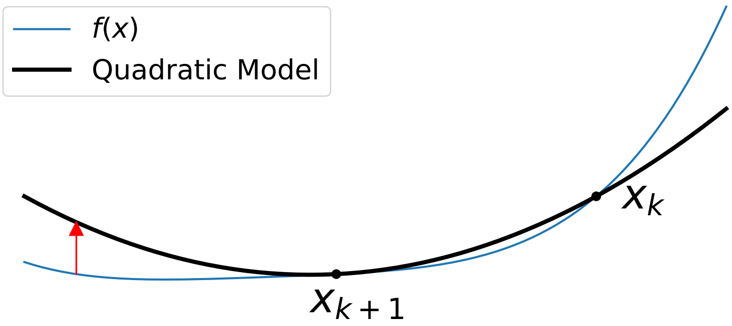


Fig. 2.5: Function-value-matching modified secant equation. The function values $f(x_k)$ and $m_k^{\mathrm{F}}(x_k)$ are matched at the previous point x_k.

By using this modified $\hat{y}_k^{\mathrm{F}}$ instead of y_k in various quasi-Newton updates, we can incorporate function value information while keeping almost the same algorithm. Additionally, since condition $s_k^\top y_k > 0$ is important for maintaining positive definiteness in BFGS update, we can consider the following modification to ensure that the inner product with s_k remains positive:

$$B_{k+1}^{\mathrm{F}'} s_k = y_k + \frac{\max(0, 2(f(x_k) - f(x_{k+1})) + (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k)}{\|s_k\|^2} s_k.$$

This is the function-value-matching modified secant condition. Refer to Figure 2.5 for an illustration.

Cubic-Augmented Modified Secant Condition

The second modification secant condition introduces a cubic term to the model, allowing simultaneous satisfaction of both the function value and the gradient matching at the previous point [54, 58, 59]. Let $T_{k+1} \in \mathbb{R}^{n\times n\times n}$ be the third-order derivative tensor of f at x_{k+1} such that

$$s_k^\top (T_{k+1} s_k) s_k = \sum_{i,j,l=1}^{n} \partial_{ijl} f(x_{k+1}) s_k^{(i)} s_k^{(j)} s_k^{(l)}.$$

Here, $\partial_{ijl} f$ denotes the third derivative of f with respect to x_i, x_j, and x_l, and $s_k^{(i)}$ is the i-th component of the vector s_k. We introduced this tensor solely for analysis purposes, and it will be eliminated from the final formula.

By incorporating this tensor term, we can define the following model:

$$\begin{aligned} m_{k+1}^{\mathrm{C}}(x) &:= f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B_{k+1}^{\mathrm{C}} (x - x_{k+1}) \\ &\quad + \frac{1}{6}(x - x_{k+1})^\top (T_{k+1}(x - x_{k+1}))(x - x_{k+1}). \end{aligned}$$

With this model, we can enforce both function value and gradient matching at the previous point x_k. Specifically, we require

$$\begin{cases} m_{k+1}^{\mathrm{C}}(x_k) = f(x_k), \\ \nabla m_{k+1}^{\mathrm{C}}(x_k) = \nabla f(x_k). \end{cases}$$

By substituting the definition of m_{k+1}^{C} and using $s_k = x_{k+1} - x_k$, we can rewrite these conditions as

$$f(x_k) = f(x_{k+1}) - s_k^\top \nabla f(x_{k+1}) + \frac{1}{2} s_k^\top B_{k+1}^{\mathrm{C}} s_k - \frac{1}{6} s_k^\top (T_{k+1} s_k) s_k,$$

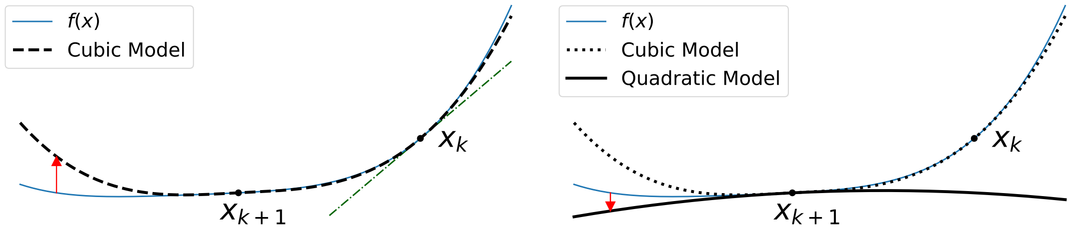


Fig. 2.6: Cubic-augmented modified secant equation. The cubic term is incorporated into the model to simultaneously match both function value and gradient at the previous point. The model is constructed using the expansion up to the quadratic term.

$$\nabla f(x_k) = \nabla f(x_{k+1}) - B^{\mathrm{C}}_{k+1} s_k + \frac{1}{2}(T_{k+1} s_k) s_k.$$

Rearranging and multiplying both sides by 3 for the first equation, and taking the inner product with s_k for the second equation, we have

$$\begin{aligned} 3(f(x_k) - f(x_{k+1})) + 3 s_k^\top \nabla f(x_{k+1}) &= \frac{3}{2} s_k^\top B^{\mathrm{C}}_{k+1} s_k - \frac{1}{2} s_k^\top (T_{k+1} s_k) s_k, \\ -s_k^\top y_k &= -s_k^\top B^{\mathrm{C}}_{k+1} s_k + \frac{1}{2} s_k^\top (T_{k+1} s_k) s_k. \end{aligned}$$

By adding these and using $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, we eliminate the tensor term and obtain the scalar identity

$$3(f(x_k) - f(x_{k+1})) + \frac{3}{2} s_k^\top (\nabla f(x_{k+1}) + \nabla f(x_k)) + \frac{1}{2} s_k^\top y_k = \frac{1}{2} s_k^\top B^{\mathrm{C}}_{k+1} s_k.$$

Again, for some scalar $\sigma^{\mathrm{C}}_k \in \mathbb{R}$, we assume $\left(B^{\mathrm{C}}_{k+1} + \sigma^{\mathrm{C}}_k I\right) s_k = y_k$. Then, the previous equation yields

$$3(f(x_k) - f(x_{k+1})) + \frac{3}{2} s_k^\top (\nabla f(x_{k+1}) + \nabla f(x_k)) = -\frac{\sigma^{\mathrm{C}}_k}{2} \|s_k\|^2,$$

which means

$$\sigma^{\mathrm{C}}_k = -\frac{6(f(x_k) - f(x_{k+1})) + 3 s_k^\top (\nabla f(x_k) + \nabla f(x_{k+1}))}{\|s_k\|^2}.$$

Thus, the modified secant condition becomes

$$B^{\mathrm{C}}_{k+1} s_k = y_k + \frac{6(f(x_k) - f(x_{k+1})) + 3 s_k^\top (\nabla f(x_k) + \nabla f(x_{k+1}))}{\|s_k\|^2} s_k.$$

This is a cubic-augmented modified secant condition. Refer to Figure 2.6 for an illustration.

2.2.2 Other Curvature Related Methods

There are several other topics related to obtaining the curvature pairs s_k, y_k. Agg-BFGS [7] is an alternative approach to managing curvature information by aggregating data rather than discarding the oldest information and adding the newest. Multi-Secant [28] extends the secant condition framework by maintaining multiple pairs of step and gradient difference vectors. In standard form we have seen, it is defined as

$$s_i = x_{i+1} - x_i, \quad y_i = \nabla f(x_{i+1}) - \nabla f(x_i). \quad (i = k - m, \dots, k)$$

In contrast, Multi-Secant centers all vectors around the latest iterate:

$$s_i = x_{k+1} - x_i, \quad y_i = \nabla f(x_{k+1}) - \nabla f(x_i). \quad (i = k-m, \ldots, k)$$

This approach can lead to better approximations in some cases.

Chapter 3

Noise-Tolerant Regularized Quasi-Newton Methods

In this chapter, we present the proposed regularized quasi-Newton method that is robust against inexact function values. This chapter is of our main contribution.

3.1 Preliminaries and Problem Setting

We begin by reviewing relevant background and formalizing the problem setting.

3.1.1 Regularized Quasi-Newton Method

The regularized quasi-Newton method is a variant of quasi-Newton methods that does not rely heavily on line search techniques. This property is important for optimization in noisy evaluation environments. There has been renewed interest in regularized Newton-type methods, including cubic-regularization schemes [37, 42, 50], and we focus on quadratic regularization [13, 27, 30, 44, 47, 48, 57].

For $g_k := \nabla f(x_k)$ and an approximate Hessian $B_k \in \mathbb{R}^{n\times n}$, regularized quasi-Newton methods compute the step as

$$x_{k+1} = x_k + d_k, \quad d_k = -(B_k + \mu_k I)^{-1} g_k,$$

with a regularization parameter $\mu_k \geq 0$. Combining regularization with line search is also possible and has been explored [47]. Several strategies for choosing μ_k in nonconvex optimization has been proposed [44, 47, 48], and one of them is given by

$$\mu_k = c_1 \max(0, -\lambda_{\min}(B_k)) + c_2 \|\nabla f(x_k)\|^{\delta},$$

where $c_1 > 1$, $c_2 > 0$, $\delta \geq 0$, and $\lambda_{\min}(B_k)$ denotes the smallest eigenvalue of B_k. This choice guarantees that $B_k + \mu_k I$ is positive definite, ensuring that d_k is a descent direction. We adapt such regularization strategies to noisy evaluation environments.

3.1.2 Objective-Function-Free Optimization

Objective-Function-Free Optimization (OFFO) refers to a class of methods that avoid explicit evaluations of the objective function and rely solely on gradient information [22, 23]. Such methods are particularly attractive when function values are unreliable due to noise. Similar adaptive trust-region approaches are also discussed [20].

Prominent examples of OFFO include adaptive gradient methods such as AdaGrad [15] and

AdaGrad-Norm [51]. Their update rules are

$$\text{(AdaGrad)} \quad x_{k+1}^{(i)} = x_k^{(i)} - \frac{1}{\theta\sqrt{\varsigma + \sum_{j=0}^{k}(g_j^{(i)})^2}} g_k^{(i)}, \qquad (i = 1, 2, \ldots, n)$$

$$\text{(AdaGrad-Norm)} \quad x_{k+1} = x_k - \frac{1}{\theta\sqrt{\varsigma + \sum_{j=0}^{k}\|g_j\|^2}} g_k$$

respectively, where $\varsigma > 0$ and $\theta > 0$ are algorithmic parameters, and $x^{(i)}$ denotes the i-th component of a vector x.

3.1.3 Problem Setting

We consider optimization problems in which exact objective function values are unavailable, following the recent literature on error-aware optimization [5, 10, 11, 21, 43, 46]. When both objective values and gradients are heavily corrupted by noise, reliable optimization cannot be guaranteed. Thus, meaningful optimization requires that at least one source of information be sufficiently accurate. This work focuses on problems arising from finite-precision computations or inexact evaluations, where objective function evaluations are subject to non-negligible errors while gradient information can be computed with relatively high accuracy. Settings in which objective values are accurate but gradients are noisy, such as derivative-free optimization or stochastic approximation methods [6, 45, 53], are beyond the scope of this work.

In the following, we formalize the assumptions on function value and gradient evaluations. We assume that only an error-contaminated function value $\overline{f}(x)$ is accessible, satisfying the hybrid absolute-relative error model [25, 26]:

$$\left|f(x) - \overline{f}(x)\right| \le \varepsilon_{\mathrm{f}} \max(1, |f(x)|), \tag{3.1}$$

where $0 \le \varepsilon_{\mathrm{f}} < 1$ is an error rate. The error is predominantly relative when $|f(x)|$ is large and effectively absolute when $|f(x)|$ is small, capturing the typical behavior of rounding errors in IEEE 754 floating-point arithmetic [25]. In the numerical experiments, ε_{f} is taken as a large multiple of the machine epsilon to account for accumulated round-off effects. Closely related models are also employed in practical stopping criteria [49].

We next specify the assumptions of gradient evaluations. In practice, termination is typically based on a gradient tolerance $\varepsilon_{\mathrm{gtol}}$, and the condition $\left\|\nabla\overline{f}(x)\right\| \le \varepsilon_{\mathrm{gtol}}$ is used as a stopping criterion. Hence, prior to termination, it holds that $\left\|\nabla\overline{f}(x)\right\| > \varepsilon_{\mathrm{gtol}}$. If the gradient noise $\left\|\nabla f(x) - \nabla\overline{f}(x)\right\|$ were comparable to or larger than $\varepsilon_{\mathrm{gtol}}$, the stopping criterion could not be satisfied reliably, even when $\nabla f(x) = 0$. To ensure meaningful termination, the gradient noise must be sufficiently small relative to $\varepsilon_{\mathrm{gtol}}$. Specifically, during the optimization process, we assume

$$\left\|\nabla f(x) - \nabla\overline{f}(x)\right\| \ll \varepsilon_{\mathrm{gtol}} < \left\|\nabla\overline{f}(x)\right\|.$$

Under this regime, the computed gradient $\nabla\overline{f}(x)$ is a sufficiently accurate approximation of the true gradient $\nabla f(x)$ prior to termination. Accordingly, for the purpose of simplifying the theoretical analysis, we adopt the exact-gradient assumption

$$\nabla\overline{f}(x) = \nabla f(x). \tag{3.2}$$

We emphasize that this assumption is introduced solely as a modeling simplification, rather than as a claim about practical gradient accuracy. Notably, the proposed algorithm does not rely on Wolfe-type conditions, which typically require accurate gradient information at trial points, making it robust to gradient inaccuracies. Our numerical experiments in section 3.5 include cases where gradients are contaminated by noise, and they also demonstrate that the proposed method remains effective even when eq. (3.2) is violated.

3.2 Proposed Algorithm

We propose a new algorithm for the problem setting stated in section 3.1.3. We draw inspiration from the OFFO framework to design robust regularization and scaling strategies. Unlike purely objective-function-free methods, our algorithm exploits function values when they are numerically reliable and smoothly switches to gradient-based control otherwise. This hybrid behavior enables improved stability without sacrificing efficiency in practical optimization tasks. The pseudo-code of our algorithm is given in Algorithm 1. The minimum of the empty set is defined as $+\infty$ in line 3. Algorithm 2 is an auxiliary line search algorithm called in Algorithm 1. In the following, we explain the three main components of Algorithm 1.

Algorithm 1: Noise Tolerant Regularized Quasi-Newton Method

Input: x_0, $0 < k_{\max}$, $0 \le \varepsilon_{\mathrm{f}} < 1$, $0 < \theta_{\min} \le \theta_{\max}$ and $0 < \varsigma < 1$

1 $k \leftarrow 0$, $g_0 \leftarrow \nabla \overline{f}(x_0)$, $K^0 \leftarrow \emptyset$, $K^+ \leftarrow \emptyset$

2 **for** $k = 0, 1, 2, \ldots, k_{\max} - 1$ **do**

3   **if** $\min_{j \in K^0} (\overline{f}(x_j) - \Delta_j) \ge \overline{f}(x_k)$ **then**

4         $K^0 \leftarrow K^0 \cup \{k\}$

5         $\mu_k \leftarrow 0$

6   **else**

7         $K^+ \leftarrow K^+ \cup \{k\}$

8         $\mu_k \leftarrow \theta_k \sqrt{\varsigma + \sum_{j \in K^+} \|g_j\|^2}$ with $\theta_k \in [\theta_{\min}, \theta_{\max}]$ (section 3.4.3).

9     $d_k \leftarrow -(B_k + \mu_k I)^{-1} g_k$ with B_k (section 3.4.1).

10   Find α_k and Δ_k satisfying eq. (3.3) by Algorithm 2.

11     $x_{k+1} \leftarrow x_k + \alpha_k d_k$, $g_{k+1} \leftarrow \nabla \overline{f}(x_{k+1})$

12 **end**

Algorithm 2: Backtracking Line Search with Relaxed Armijo Condition

Input: $x_k \in \mathbb{R}^n$, $d_k \in \mathbb{R}^n$, $g_k \in \mathbb{R}^n$, $0 < c < 1$ and $0 < \beta_{\min} < \beta_{\max} < 1$

$\alpha_k \leftarrow 1$

$\Delta_k \leftarrow \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \max(1, \overline{f}(x_k), -\overline{f}(x_k + \alpha_k d_k))$

while $\overline{f}(x_k) + c\alpha_k g_k^\top d_k + \Delta_k < \overline{f}(x_k + \alpha_k d_k)$ **do**

    $\alpha_k \leftarrow \beta \alpha_k$ with $\beta \in [\beta_{\min}, \beta_{\max}]$ (section 3.4.2).

    $\Delta_k \leftarrow \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \max(1, \overline{f}(x_k), -\overline{f}(x_k + \alpha_k d_k))$

end

return α_k

Computation of the Regularized Quasi-Newton Direction (line 9): The first component is the computation of the regularized quasi-Newton direction d_k stated in section 3.1.1. Given $g_k = \nabla \overline{f}(x_k) = \nabla f(x_k)$, an approximate Hessian B_k, and a regularization parameter μ_k, we compute the search direction $d_k = -(B_k + \mu_k I)^{-1} g_k$. A practical choice of B_k is discussed in section 3.4.1.

Relaxed Armijo Condition with an Error-Absorbing Term (line 10): The second component is the computation of the step size α_k using a relaxed Armijo condition, which is defined as follows:

$$\begin{cases} \overline{f}(x_k) + c\alpha_k g_k^\top d_k + \Delta_k \geq \overline{f}(x_k + \alpha_k d_k), \\ \Delta_k = \dfrac{2\varepsilon_\mathrm{f}}{1-\varepsilon_\mathrm{f}} \max(1, \overline{f}(x_k), -\overline{f}(x_k + \alpha_k d_k)), \end{cases} \tag{3.3}$$

where Δ_k is the error-absorbing term that is recomputed for each α_k. This relaxation guarantees the existence of $\alpha_k \in (0,1]$ even in the presence of numerical errors in $\overline{f}$. When d_k is a descent direction ($g_k^\top d_k < 0$), eq. (3.3) ensures that a temporary increase of $\overline{f}$ is at most Δ_k. Since Δ_k depends on $-\overline{f}(x_k + \alpha_k d_k)$ but not on $\overline{f}(x_k + \alpha_k d_k)$, Δ_k does not increase when $\overline{f}(x_k + \alpha_k d_k)$ becomes large. This prevents unbounded growth of Δ_k.

Adaptive Update of the Regularization Parameter (lines 3–8): The third component is the adaptive update of the regularization parameter μ_k. A larger μ_k yields a more conservative step, while a smaller μ_k allows a more aggressive quasi-Newton step, especially when $\mu_k = 0$. In practical settings, setting $\mu_k = 0$ is preferred, since this allows us to fully exploit the quasi-Newton approximation and typically leads to practically faster convergence. Let us partition the iteration indices $K := \{0, 1, 2, \ldots, k_{\max} - 1\}$ into two sets:

$$K^0 := \{k \in K \mid \mu_k = 0\}, \quad K^+ := \{k \in K \mid \mu_k > 0\}. \tag{3.4}$$

We set $\mu_k = 0$ if and only if the following condition is satisfied:

$$\min_{j \in K^0, j<k} \left(\overline{f}(x_j) - \Delta_j\right) \geq \overline{f}(x_k). \tag{3.5}$$

Note that $k = 0$ satisfies eq. (3.5) since K^0 is empty. This condition restricts $\mu_k = 0$ to iterations for which a sufficient decrease in $\overline{f}$ has been observed. Thereby, while $\overline{f}$ can temporarily increase, this condition theoretically ensures that $\overline{f}$ does not grow unboundedly or oscillate indefinitely.

Otherwise, to ensure convergence to first-order stationary points, we apply an OFFO-based update

$$\mu_k = \theta_k \sqrt{\varsigma + \sum_{j \in K^+, j \leq k} \|g_j\|^2}, \tag{3.6}$$

as described in section 3.1.2. Since OFFO schemes do not rely on function values, this update is robust to numerical errors in $\overline{f}$.

3.3 Theoretical Analysis

In this section, we establish the global convergence properties of Algorithm 1 under the problem setting in section 3.1.3, namely, eqs. (3.1) and (3.2).

3.3.1 Assumptions

Throughout this section, we make the following three assumptions. The first assumption ensures that the objective function is bounded below.

Assumption 1. *The function f is bounded below, meaning that for all $x \in \mathbb{R}^n$,*

$$f(x) \geq f^* := \inf_{x \in \mathbb{R}^n} f(x) > -\infty.$$

By the triangle inequality and eq. (3.1), we have $\overline{f}(x) \geq f(x) - \varepsilon_{\mathrm{f}} \max(1, |f(x)|)$. Since the function $g(t) = t - \varepsilon_{\mathrm{f}} \max(1, |t|)$ is non-decreasing for $0 \leq \varepsilon_{\mathrm{f}} < 1$, it follows that $\inf_x g(f(x)) = g(f^*)$. Consequently, we can also lower bound $\overline{f}$ as

$$\overline{f}(x) \geq \overline{f}^* := \inf_{x \in \mathbb{R}^n} (f(x) - \varepsilon_{\mathrm{f}} \max(1, |f(x)|)) = f^* - \varepsilon_{\mathrm{f}} \max(1, |f^*|) > -\infty. \tag{3.7}$$

The second assumption ensures smoothness of the objective function.

Assumption 2. *The function $f : \mathbb{R}^n \to \mathbb{R}$ is L-smooth, meaning that there exists a constant $L > 0$ such that for all $x, y \in \mathbb{R}^n$,*

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2.$$

The third assumption requires the matrices $\{B_k\}_{k=0}^{k_{\max}-1}$ to be uniformly positive definite and bounded.

Assumption 3. *There exist constants $0 < m \leq M$ such that all B_k satisfy*

$$mI \preceq B_k \preceq MI.$$

As we will discuss in section 3.4, the standard BFGS update with a slight modification guarantees Assumption 3 in practice. By Assumption 3, we can derive useful bounds involving g_k and d_k. Using $d_k = -(B_k + \mu_k I)^{-1} g_k$ (line 9) and Assumption 3 asserting $(m + \mu_k)I \preceq B_k + \mu_k I \preceq (M + \mu_k)I$, we obtain the following bounds:

$$-g_k^\top d_k = d_k^\top (B_k + \mu_k I) d_k \geq (m + \mu_k) \|d_k\|^2 \geq m \|d_k\|^2, \tag{3.8}$$

$$-g_k^\top d_k = g_k^\top (B_k + \mu_k I)^{-1} g_k \geq \frac{1}{M + \mu_k} \|g_k\|^2, \tag{3.9}$$

$$\|d_k\|^2 = \left\|(B_k + \mu_k I)^{-1} g_k\right\|^2 \leq \frac{1}{(m + \mu_k)^2} \|g_k\|^2. \tag{3.10}$$

3.3.2 Error Bound on Function Evaluations

We first establish useful properties of the error-absorbing term Δ_k defined in eq. (3.3). In the following, we define the actual and observed function decreases as

$$\delta_k := f(x_k) - f(x_{k+1}), \tag{3.11}$$

$$\overline{\delta}_k := \overline{f}(x_k) - \overline{f}(x_{k+1}). \tag{3.12}$$

The first property states that the error between the actual values difference δ_k and the observed values difference $\overline{\delta}_k$ can be bounded by Δ_k.

Lemma 1. *If x_k and x_{k+1} satisfy $f(x_{k+1}) \leq f(x_k)$, it holds that*

$$\delta_k \leq \overline{\delta}_k + \Delta_k. \tag{3.13}$$

Proof. We start from general observations. We first prove that, for all $x \in \mathbb{R}^n$,

$$\begin{cases} \max(1, +f(x)) \leq \frac{1}{1-\varepsilon_{\mathrm{f}}} \max(1, +\overline{f}(x)), \\ \max(1, -f(x)) \leq \frac{1}{1-\varepsilon_{\mathrm{f}}} \max(1, -\overline{f}(x)). \end{cases} \tag{3.14}$$

We prove the inequality with the plus sign. If $+f(x) \leq 1$, then $\max(1, +f(x)) = 1 \leq \frac{1}{1-\varepsilon_\mathrm{f}} \leq \frac{1}{1-\varepsilon_\mathrm{f}} \max(1, +\overline{f}(x))$. If $+f(x) > 1$, eq. (3.1) implies

$$\left|f(x) - \overline{f}(x)\right| \leq \varepsilon_\mathrm{f} \max(1, |f(x)|) = +\varepsilon_\mathrm{f} f(x) \tag{3.15}$$

and thus $f(x) - \overline{f}(x) \leq +\varepsilon_\mathrm{f} f(x)$. Hence,

$$\max(1, +f(x)) = +f(x) \leq +\frac{1}{1-\varepsilon_\mathrm{f}} \overline{f}(x) \leq \frac{1}{1-\varepsilon_\mathrm{f}} \max(1, +\overline{f}(x)).$$

The case with the negative sign can be shown analogously, using $-f(x) + \overline{f}(x) \leq -\varepsilon_\mathrm{f} f(x)$, which follows from eq. (3.15). This completes the proof of eq. (3.14).

We also have the following general bound for all $k \in K$:

$$\begin{aligned}
\left|\delta_k - \overline{\delta}_k\right| &= \left|(f(x_k) - \overline{f}(x_k)) - (f(x_{k+1}) - \overline{f}(x_{k+1}))\right| && \text{(rearrangement)} \\
&\leq \left|f(x_k) - \overline{f}(x_k)\right| + \left|f(x_{k+1}) - \overline{f}(x_{k+1})\right| && \text{(triangle inequality)} \\
&\leq \varepsilon_\mathrm{f} \max(1, |f(x_k)|) + \varepsilon_\mathrm{f} \max(1, |f(x_{k+1})|) && \text{(eq. (3.1))} \\
&\leq 2\varepsilon_\mathrm{f} \max(1, |f(x_k)|, |f(x_{k+1})|). && \text{(max property)}
\end{aligned} \tag{3.16}$$

With these preparations, we now prove the lemma. From the condition $f(x_{k+1}) \leq f(x_k)$, eq. (3.16) can be further bounded as

$$\begin{aligned}
2\varepsilon_\mathrm{f} \max(1, |f(x_k)|, |f(x_{k+1})|) &= 2\varepsilon_\mathrm{f} \max(1, f(x_k), -f(x_{k+1})) && (f(x_{k+1}) \leq f(x_k)) \\
&\leq \frac{2\varepsilon_\mathrm{f}}{1-\varepsilon_\mathrm{f}} \max(1, \overline{f}(x_k), -\overline{f}(x_{k+1})) && \text{(eq. (3.14))} \\
&= \Delta_k, && \text{(eq. (3.3))}
\end{aligned}$$

which yields the desired inequality with $\delta_k - \overline{\delta}_k \leq \left|\delta_k - \overline{\delta}_k\right| \leq \Delta_k$. □

We also derive the following lemma for later use.

Lemma 2. *Under Assumption 3, it holds that*

$$\overline{\delta}_k \leq \delta_k + \frac{1+\varepsilon_\mathrm{f}}{1-\varepsilon_\mathrm{f}} \Delta_k.$$

Proof. Since $-g_k^\top d_k \geq 0$ from Assumption 3 and eq. (3.8), the relaxed Armijo condition (eq. (3.3)) implies

$$\overline{f}(x_k) - \overline{f}(x_{k+1}) \geq -c\alpha_k g_k^\top d_k - \Delta_k \geq -\Delta_k. \tag{3.17}$$

Thus, using the general results in eq. (3.16) and $\Delta_k \geq 0$, we can further bound as

$$\begin{aligned}
\left|\delta_k - \overline{\delta}_k\right| &\leq 2\varepsilon_\mathrm{f} \max(1, |f(x_k)|, |f(x_{k+1})|) && \text{(eq. (3.16))} \\
&\leq \frac{2\varepsilon_\mathrm{f}}{1-\varepsilon_\mathrm{f}} \max(1, \overline{f}(x_k), -\overline{f}(x_k), \overline{f}(x_{k+1}), -\overline{f}(x_{k+1})) && \text{(eq. (3.14))} \\
&\leq \frac{2\varepsilon_\mathrm{f}}{1-\varepsilon_\mathrm{f}} \max(1, \overline{f}(x_k) + \Delta_k, -\overline{f}(x_{k+1}) + \Delta_k) && \text{(eq. (3.17))} \\
&\leq \frac{2\varepsilon_\mathrm{f}}{1-\varepsilon_\mathrm{f}} \left(\max(1, \overline{f}(x_k), -\overline{f}(x_{k+1})) + \Delta_k\right) \\
&= \left(1 + \frac{2\varepsilon_\mathrm{f}}{1-\varepsilon_\mathrm{f}}\right) \Delta_k = \frac{1+\varepsilon_\mathrm{f}}{1-\varepsilon_\mathrm{f}} \Delta_k, && \text{(eq. (3.3))}
\end{aligned}$$

which yields the desired inequality with $\overline{\delta}_k - \delta_k \leq \left|\delta_k - \overline{\delta}_k\right| \leq \frac{1+\varepsilon_\mathrm{f}}{1-\varepsilon_\mathrm{f}} \Delta_k$. □

3.3.3 Backtracking Stage

We next analyze the line search in Algorithm 2. Recall that the parameter c in Algorithm 2 satisfy $0 < c < 1$ and $0 < m$ from Assumption 3.

Lemma 3. *Under Assumptions 2 and 3, the backtracking procedure in Algorithm 2 terminates whenever the step size α_k satisfies*

$$\alpha_k \leq \frac{2(1-c)m}{L}. \tag{3.18}$$

Proof. Under the L-smoothness of f (Assumption 2), we always have

$$\begin{aligned} f(x_k) - f(x_k + \alpha_k d_k) &\geq -\alpha_k g_k^\top d_k - \frac{L}{2}\alpha_k^2 \|d_k\|^2 \\ &= -c\alpha_k g_k^\top d_k + \alpha_k \left(-(1-c) g_k^\top d_k - \frac{L\alpha_k}{2} \|d_k\|^2 \right). \end{aligned} \tag{3.19}$$

From Assumption 3 and eq. (3.8), we have $-g_k^\top d_k \geq m\|d_k\|^2$. Under eq. (3.18), we have

$$-(1-c) g_k^\top d_k - \frac{L\alpha_k}{2}\|d_k\|^2 \geq \left((1-c)m - \frac{L\alpha_k}{2} \right) \|d_k\|^2 \geq 0. \tag{3.20}$$

Combining eqs. (3.19) and (3.20) yields

$$f(x_k) - f(x_k + \alpha_k d_k) \geq -c\alpha_k g_k^\top d_k.$$

Since $-g_k^\top d_k \geq m\|d_k\|^2 \geq 0$ from Assumption 3 and eq. (3.8), the condition of Lemma 1 is satisfied with $x_{k+1} = x_k + \alpha_k d_k$. Thus we obtain $\delta_k \leq \overline{\delta}_k + \Delta_k$, i.e.,

$$-c\alpha_k g_k^\top d_k \leq f(x_k) - f(x_k + \alpha_k d_k) \leq \overline{f}(x_k) - \overline{f}(x_k + \alpha_k d_k) + \Delta_k.$$

This inequality means that the relaxed Armijo condition (eq. (3.3)) is satisfied and thus the backtracking procedure terminates. This completes the proof. □

Now, let us analyze Algorithm 2 using Lemma 3. Define

$$\overline{\alpha} := \min\left(\beta_{\min} \frac{2(1-c)m}{L}, 1 \right) > 0. \tag{3.21}$$

The following states the key property of the backtracking procedure.

Proposition 1. *Under Assumptions 2 and 3, the backtracking line search in Algorithm 2 terminates after finitely many rejections, and the step size α_k satisfies*

$$\alpha_k \geq \overline{\alpha}.$$

Proof. The line 4 in Algorithm 2 multiplies α_k by β at each rejection, satisfying $0 < \beta_{\min} \leq \beta \leq \beta_{\max} < 1$. For the number of backtracking rejections r, we have $\alpha_k \leq \beta_{\max}^r$. Thus, r is finite since the backtracking procedure terminates for sufficiently small α_k by Lemma 3. If $r = 0$, $\alpha_k = 1 \geq \overline{\alpha}$. Otherwise, $\alpha_k \geq \beta_{\min}\alpha_k'$ where α_k' is the last rejected step size. By Lemma 3, we have $\alpha_k' > \frac{2(1-c)m}{L}$, and thus

$$\alpha_k \geq \beta_{\min}\alpha_k' > \beta_{\min}\frac{2(1-c)m}{L} \geq \overline{\alpha}. \tag{3.22}$$

This completes the proof. □

3.3.4 Bound for the case $\mu_k = 0$

In this subsection, we lower bound the sum of actual function decreases δ_k over $k \in K^0$. Recall that K^0 denotes the set of iteration indices k such that $\mu_k = 0$, as in eq. (3.4). To this end, we establish that the sum of Δ_k over $k \in K^0$ is upper bounded, where Δ_k is the error-absorbing term in the relaxed Armijo condition (eq. (3.3)). We introduce the following constant $\Delta_{\max}$, which serves as an upper bound for Δ_k at $k \in K^0$.

$$\Delta_{\max} := \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \max(1, \overline{f}(x_0), -\overline{f}^*). \tag{3.23}$$

Lemma 4. *Under Assumptions 1 to 3, the sum of Δ_k over $k \in K^0$ satisfies*

$$\sum_{k \in K^0} \Delta_k \leq \overline{f}(x_0) - \overline{f}^* + \Delta_{\max}.$$

Proof. Let $(k_i)_{i=0}^{\ell}$ with $\ell \geq 0$ be the elements of K^0 in increasing order. We have $\overline{f}(x_{k_\ell}) \leq \overline{f}(x_0)$ from eq. (3.5) with $k = k_\ell$, where the case $\ell = 0$ follows from $k_0 = 0$. Combining this with Assumption 1, we have $\Delta_{k_\ell} \leq \Delta_{\max}$. For all $i < \ell$, eq. (3.5) with $j = k_i$ and $k = k_{i+1}$ yields

$$\Delta_{k_i} \leq \overline{f}(x_{k_i}) - \overline{f}(x_{k_{i+1}}), \tag{3.24}$$

and summing Δ_k over $k \in K^0$ gives

$$\begin{aligned}
\sum_{k \in K^0} \Delta_k &\leq \sum_{i=0}^{\ell-1} (\overline{f}(x_{k_i}) - \overline{f}(x_{k_{i+1}})) + \Delta_{k_\ell} && \text{(eq. (3.24))} \\
&= \overline{f}(x_0) - \overline{f}(x_{k_\ell}) + \Delta_{k_\ell} && \text{(telescoping sum)} \\
&\leq \overline{f}(x_0) - \overline{f}^* + \Delta_{\max}. && (-\overline{f}(x_{k_\ell}) \leq -\overline{f}^* \text{ and } \Delta_{k_\ell} \leq \Delta_{\max})
\end{aligned}$$

This completes the proof. □

We now bound the sum of $\delta_k = f(x_k) - f(x_{k+1})$ over $k \in K^0$ with $\|g_k\|^2$ as follows.

Lemma 5. *Under Assumptions 1 to 3, the sum of δ_k over $k \in K^0$ satisfies*

$$\sum_{k \in K^0} \delta_k \geq \frac{c\overline{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 - \frac{2}{1-\varepsilon_{\mathrm{f}}} \left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max}\right).$$

Proof. The relaxed Armijo condition (eq. (3.3)) at iteration $k \in K^0$ yields

$$\begin{aligned}
\overline{\delta}_k + \Delta_k &= \overline{f}(x_k) - \overline{f}(x_{k+1}) + \Delta_k && \text{(eq. (3.12))} \\
&\geq -c\alpha_k g_k^\top d_k && \text{(eq. (3.3))} \\
&\geq \frac{c\overline{\alpha}}{M + \mu_k} \|g_k\|^2 && \text{(Proposition 1 and eq. (3.9))} \\
&= \frac{c\overline{\alpha}}{M} \|g_k\|^2. && (\mu_k = 0 \text{ since } k \in K^0) \qquad (3.25)
\end{aligned}$$

Using Lemma 2, we also have

$$\overline{\delta}_k + \Delta_k \leq \delta_k + \frac{1+\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \Delta_k + \Delta_k = \delta_k + \frac{2}{1-\varepsilon_{\mathrm{f}}} \Delta_k. \tag{3.26}$$

Then, combining eqs. (3.25) and (3.26) and summing over $k \in K^0$ gives

$$\sum_{k\in K^0} \frac{c\overline{\alpha}}{M}\|g_k\|^2 \leq \sum_{k\in K^0} \delta_k + \frac{2}{1-\varepsilon_{\mathrm{f}}}\sum_{k\in K^0}\Delta_k \leq \sum_{k\in K^0}\delta_k + \frac{2}{1-\varepsilon_{\mathrm{f}}}\left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max}\right),$$

where the last inequality follows from Lemma 4. Rearranging this completes the proof. □

3.3.5 Bound for the case $\mu_k > 0$

Next, we show that the sum of δ_k over $k \in K^+$ is lower bounded. Recall that K^+ be a set of iteration indices k such that $\mu_k > 0$ as defined in eq. (3.4), and that μ_k is defined as $\theta_k\sqrt{\varsigma + \sum_{j\in K^+,\ j<k}\|g_j\|^2}$ with $\theta_k \in [\theta_{\min}, \theta_{\max}]$ in line 8 of Algorithm 1.

Before proceeding to the main proof, we present Lemma 6, providing standard inequalities in the AdaGrad-Norm algorithm analysis [51, Lemma 3.2], [23, Lemma 3.1]. Its proof is deferred to section A.1.

Lemma 6. *In Algorithm 1, we have*

$$\sum_{k\in K^+}\frac{\|g_k\|^2}{\mu_k^2} \leq \frac{1}{\theta_{\min}^2}\left(\ln\left(\varsigma + \sum_{k\in K^+}\|g_k\|^2\right) - \ln(\varsigma)\right),$$
$$\sum_{k\in K^+}\frac{\|g_k\|^2}{\mu_k} \geq \frac{1}{\theta_{\max}}\left(\sqrt{\varsigma + \sum_{k\in K^+}\|g_k\|^2} - \sqrt{\varsigma}\right).$$

We now lower bound the sum of δ_k over $k \in K^+$ with $\|g_k\|^2$. We define the following constants for later use:

$$C_1 := \frac{\overline{\alpha}}{\theta_{\max}}, \qquad C_2 := \frac{M\overline{\alpha} + L/2}{\theta_{\min}^2}. \tag{3.27}$$

Lemma 7. *Under Assumptions 2 and 3, the sum of δ_k over $k \in K^+$ satisfies*

$$\sum_{k\in K^+}\delta_k \geq C_1\left(\sqrt{\varsigma + \sum_{k\in K^+}\|g_k\|^2} - \sqrt{\varsigma}\right) - C_2\left(\ln\left(\varsigma + \sum_{k\in K^+}\|g_k\|^2\right) - \ln(\varsigma)\right).$$

Proof. From the L-smoothness (Assumption 2), for $k \in K^+$ we have

$$\begin{aligned}
\delta_k &= f(x_k) - f(x_k + \alpha_k d_k) && \text{(eq. (3.11) and } x_{k+1} = x_k + \alpha_k d_k) \\
&\geq -\alpha_k g_k^\top d_k - \frac{\alpha_k^2 L}{2}\|d_k\|^2 && \text{(Assumption 2)} \\
&\geq \left(\frac{\alpha_k}{M+\mu_k} - \frac{\alpha_k^2 L}{2(m+\mu_k)^2}\right)\|g_k\|^2 && \text{(eqs. (3.9) and (3.10))} \\
&\geq \left(\frac{\overline{\alpha}}{M+\mu_k} - \frac{L}{2(m+\mu_k)^2}\right)\|g_k\|^2. && \text{(Proposition 1 and } \alpha_k \leq 1) \qquad (3.28)
\end{aligned}$$

We can further bound the terms in eq. (3.28) as

$$\frac{\overline{\alpha}}{M+\mu_k} = \frac{\overline{\alpha}}{\mu_k}\frac{1}{1+\frac{M}{\mu_k}} \geq \frac{\overline{\alpha}}{\mu_k}\left(1 - \frac{M}{\mu_k}\right) = \frac{\overline{\alpha}}{\mu_k} - \frac{M\overline{\alpha}}{\mu_k^2}, \qquad \frac{L}{2(m+\mu_k)^2} \leq \frac{L}{2\mu_k^2}. \tag{3.29}$$

Thus, applying eq. (3.29) to eq. (3.28) yields

$$\delta_k \geq \left(\frac{\overline{\alpha}}{\mu_k} - \frac{M\overline{\alpha} + L/2}{\mu_k^2}\right)\|g_k\|^2. \tag{3.30}$$

Summing eq. (3.30) over $k \in K^+$ and applying Lemma 6 yield

$$\begin{aligned}\sum_{k\in K^+} \delta_k &\geq \sum_{k\in K^+}\left(\frac{\overline{\alpha}}{\mu_k} - \frac{M\overline{\alpha} + L/2}{\mu_k^2}\right)\|g_k\|^2 \\ &\geq \frac{\overline{\alpha}}{\theta_{\max}}\left(\sqrt{\varsigma + \sum_{k\in K^+}\|g_k\|^2} - \sqrt{\varsigma}\right) - \frac{M\overline{\alpha} + L/2}{\theta_{\min}^2}\left(\ln\left(\varsigma + \sum_{k\in K^+}\|g_k\|^2\right) - \ln(\varsigma)\right),\end{aligned}$$

which yields the desired result. □

3.3.6 Global Convergence Result

Finally, we combine the results for $k \in K^0$ (Lemma 5) and $k \in K^+$ (Lemma 7) to derive an upper bound on the total sum of $\|g_k\|^2$. We define the constant F as follows:

$$F := f(x_0) - f^* + \frac{2}{1-\varepsilon_{\mathrm{f}}}\left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max}\right) + C_1\sqrt{\varsigma} - C_2\ln(\varsigma). \tag{3.31}$$

Then, we can obtain the following unified bound of the sum of $\|g_k\|^2$.

Lemma 8. *Under the assumptions, the sum of $\|g_k\|^2$ over $k \in K^0, K^+$ satisfies*

$$F \geq \frac{c\overline{\alpha}}{M}\sum_{k\in K^0}\|g_k\|^2 + C_1\sqrt{\varsigma + \sum_{k\in K^+}\|g_k\|^2} - C_2\ln\left(\varsigma + \sum_{k\in K^+}\|g_k\|^2\right),$$

Proof. Since $f(x_{k_{\max}}) \geq f^*$ by Assumption 1 and $K = K^0 \cup K^+$, we have

$$f(x_0) - f^* \geq f(x_0) - f(x_{k_{\max}}) = \sum_{k\in K}\delta_k = \sum_{k\in K^0}\delta_k + \sum_{k\in K^+}\delta_k.$$

From Lemmas 5 and 7, we can further bound as

$$\begin{aligned}f(x_0) - f^* \geq \frac{c\overline{\alpha}}{M}\sum_{k\in K^0}\|g_k\|^2 &- \frac{2}{1-\varepsilon_{\mathrm{f}}}\left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max}\right) \\ &+ C_1\sqrt{\varsigma + \sum_{k\in K^+}\|g_k\|^2} - C_1\sqrt{\varsigma} - C_2\ln\left(\varsigma + \sum_{k\in K^+}\|g_k\|^2\right) + C_2\ln(\varsigma).\end{aligned}$$

Substituting eq. (3.31) yields the desired bound, concluding the proof. □

Now, we establish the following global convergence result for Algorithm 1.

Theorem 1 (Global Convergence)**.** *Under Assumptions 1 to 3, the average of squared gradient norms generated by Algorithm 1 satisfies:*

$$\frac{1}{k_{\max}}\sum_{k=0}^{k_{\max}-1}\|g_k\|^2 = \mathcal{O}\left(\frac{1}{k_{\max}}\right).$$

Proof. Let us define the function $\phi : \mathbb{R}_{>0} \to \mathbb{R}$ as

$$\phi(G) := \frac{C_1}{2}\sqrt{G} - C_2 \ln(G). \tag{3.32}$$

Since $\phi'(G) = \frac{C_1}{4\sqrt{G}} - \frac{C_2}{G}$, its minimum over $G > 0$ is attained at $G^* := \left(\frac{4C_2}{C_1}\right)^2$ with

$$\phi(G^*) = 2C_2\left(1 - \ln\left(\frac{4C_2}{C_1}\right)\right) = 2C_2 \ln\left(\frac{\mathrm{e}C_1}{4C_2}\right),$$

where e is the base of the natural logarithm. Thus, Lemma 8 yields

$$\begin{aligned} F &\geq \frac{c\overline{\alpha}}{M} \sum_{k\in K^0} \|g_k\|^2 + \frac{C_1}{2}\sqrt{\sum_{k\in K^+} \|g_k\|^2} + \phi\left(\sum_{k\in K^+} \|g_k\|^2 + \varsigma\right) \\ &\geq \frac{c\overline{\alpha}}{M} \sum_{k\in K^0} \|g_k\|^2 + \frac{C_1}{2}\sqrt{\sum_{k\in K^+} \|g_k\|^2} + \phi(G^*), \end{aligned}$$

which is equivalent to

$$0 \leq \frac{c\overline{\alpha}}{M} \sum_{k\in K^0} \|g_k\|^2 + \frac{C_1}{2}\sqrt{\sum_{k\in K^+} \|g_k\|^2} \leq F' \tag{3.33}$$

where

$$\begin{aligned} F' &:= F - \phi(G^*) \\ &= f(x_0) - f^* + \frac{2}{1-\varepsilon_{\mathrm{f}}}\left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max}\right) + C_1\sqrt{\varsigma} - C_2\ln(\varsigma) - 2C_2\ln\left(\frac{\mathrm{e}C_1}{4C_2}\right). \end{aligned}$$

It remains to bound $\sum_{k\in K} \|g_k\|^2 = \sum_{k\in K^0} \|g_k\|^2 + \sum_{k\in K^+} \|g_k\|^2$ given the constraint in eq. (3.33). More generally, let $a, b > 0$, $c \geq 0$ and $x, y \geq 0$, and consider the maximization problem $x + y$ subject to $ax + b\sqrt{y} \leq c$. The maximum is achieved on the boundary $ax + b\sqrt{y} = c$ since $x + y$ is increasing in both variables. On this boundary, we can write $y = \left(\frac{c-ax}{b}\right)^2$, and thus $x + y$ is convex on an interval $0 \leq x \leq c/a$, attaining its maximum at one of the endpoints. Evaluating at the endpoints yields the candidates $(x, y) = (c/a, 0)$ and $(x, y) = (0, (c/b)^2)$, and therefore the maximum of $x + y$ is $\max\left(c/a, (c/b)^2\right)$. Applying this result to eq. (3.33) gives

$$\begin{aligned} \frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 &= \frac{1}{k_{\max}}\left(\sum_{k\in K^0} \|g_k\|^2 + \sum_{k\in K^+} \|g_k\|^2\right) \qquad (K = K^0 \cup K^+) \\ &\leq \frac{1}{k_{\max}} \max\left(\frac{MF'}{c\overline{\alpha}}, \frac{4F'^2}{C_1^2}\right), \end{aligned} \tag{3.34}$$

which completes the proof. □

As a remark, the function $\phi(G)$ in eq. (3.32) is closely related to the Lambert W function [12, 23]. For $x \in [-1/\mathrm{e}, 0)$, the equation $ye^y = x$ admits two real negative solutions, and the smaller one is given by the second branch of Lambert W function $y = W_{-1}(x) < 0$. Using this function, the equation $\phi(G) = 0$ can be solved in closed form, which leads to a sharper bound in Theorem 1. Since this is not essential for the main argument, we omit the details and refer to [23].

We next interpret eq. (3.34) in the proof of Theorem 1. The contribution of the indices in K^0 is analogous to the classical analysis of gradient descent. Indeed, for gradient descent with fixed step size $\alpha = 1/L$, one has

$$\frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 \leq= \frac{1}{k_{\max}} \frac{2(f(x_0) - f^*)}{\alpha}.$$

Thus, the term associated with K^0 reflects the standard $\mathcal{O}(f(x_0) - f^*)$ dependence of first-order methods since F' contains the initial gaps $f(x_0) - f^*$ and $\overline{f}(x_0) - \overline{f}^*$. In contrast, the contribution of K^+ resembles the behavior of AdaGrad-Norm, whose convergence bound scales quadratically with the initial optimality gap [51].

Finally, it is well-known that Theorem 1 implies that the iteration complexity:

$$\min\{k \mid \|\nabla f(x_k)\| \le \varepsilon_{\text{gtol}}\} = \mathcal{O}(1/\varepsilon_{\text{gtol}}^2).$$

This is a standard convergence guarantee for first-order optimization algorithms applied to L-smooth nonconvex functions.

3.4 Practical Choices of Algorithmic Variables

In sections 3.2 and 3.3, we introduced Algorithm 1 and established its convergence properties. In this section, we discuss practical choices of algorithmic variables and parameters that lead to fast and stable performance. The quantities to be discussed here are the matrices B_k, the step sizes α_k, and the regularization parameters μ_k.

3.4.1 Approximate Hessian B_k

We first discuss the construction of the matrices B_k used in line 9 of Algorithm 1. The purpose of this subsection is to show that B_k can satisfy Assumption 3, the uniform spectral bound, with a slight modification and selection of curvature pairs using the L-BFGS method.

Construction of B_k: We briefly note the L-BFGS method and the curvature pair notation [7, 31, 38]. For an iteration k, we define

$$s_k := x_{k+1} - x_k, \qquad y_k := g_{k+1} - g_k \tag{3.35}$$

where g_k and g_{k+1} are the gradients at x_k and x_{k+1}, respectively. Starting from an initial matrix, an approximate Hessian is obtained by successively applying BFGS updates to a sequence of curvature pairs. Let $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$ be a collection of curvature pairs, where k_i denotes the iteration index at which the i-th pair was generated. Starting from the initial matrix γI with $\gamma = \|y_{k_0}\|^2 / y_{k_0}^\top s_{k_0}$, we define $\mathrm{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1})$ as the matrix B_k obtained by sequentially applying the BFGS update using these pairs in the given order. Here, p denotes the maximum number of stored curvature pairs and is a fixed small integer in the L-BFGS method. When fewer than p curvature pairs are available, the BFGS update is applied using all currently available pairs.

Now, we provide a sufficient condition for Assumption 3 by the following proposition, which is a variant of existing work [17, Theorem 5.5] [19, Lemma 1]. Its proof is deferred to section A.2.

Proposition 2. *Let $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$ be a fixed number of curvature pairs with $s_{k_i} \neq 0$. Assume that there exist constants $0 < \lambda \le \Lambda$ such that for all $(s, y) \in \{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$,*

$$y^\top s \ge \frac{1}{\Lambda}\|y\|^2, \tag{3.36}$$

$$y^\top s \ge \lambda \|s\|^2. \tag{3.37}$$

Define the condition number $\kappa := \Lambda/\lambda$. Then there exist $0 < m < M$ such that

$$mI \preceq \mathrm{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1}) \preceq MI, \tag{3.38}$$

where

$$M := \gamma + p\Lambda, \qquad m := \left((1+\sqrt{\kappa})^{2p}\left(\gamma + \frac{1}{\lambda(2\sqrt{\kappa}+\kappa)}\right)\right)^{-1}.$$

Proposition 2 shows that, whenever eqs. (3.36) and (3.37) hold for the stored pairs, the resulting BFGS matrix satisfies Assumption 3 as expressed in eq. (3.38).

Based on Proposition 2, we basically adopt the following rule for forming B_k:

$$B_k = \mathrm{BFGS}(\{(s_{k_i}, \bar{y}_{k_i})\}), \tag{3.39}$$

where

$$\bar{y}_{k_i} := \theta_i^{\mathrm{damp}} y_{k_i} + (1 - \theta_i^{\mathrm{damp}}) B_{k-1} s_{k_i}$$

with $\theta_i^{\mathrm{damp}} \in [0, 1]$. This parameter is determined using a damped BFGS update [32, 41] and the indices $\{k_i\}_{i=0}^{p-1}$ are selected so that the modified pairs $(s_{k_i}, \bar{y}_{k_i})$ satisfy eqs. (3.36) and (3.37) for some constants λ, Λ. The damped BFGS is a classical technique to ensure $\bar{y}_{k_i}^\top s_{k_i} > 0$, and we adopted its simple limited-memory variant [32].

Since we do not employ Wolfe conditions in the line search, this damping technique plays a central role in ensuring the positive definiteness of B_k. The selection of curvature pairs shares the same spirit as the cautious update [29, 33, 34], which only uses pairs satisfying certain curvature conditions. By Proposition 2, B_k satisfies the uniform spectral bound in Assumption 3. This confirms that the practical construction of B_k is consistent with the theoretical requirements.

Remarks for Practical Implementation: Having established that the proposed construction of B_k satisfies Assumption 3, we now present several remarks on its practical implementation. We begin by describing how to compute the search direction

$$d_k = -(B_k + \mu_k I)^{-1} g_k \tag{3.40}$$

with B_k in eq. (3.39). When $\mu_k = 0$, the direction can be computed efficiently by the standard L-BFGS two-loop recursion [31, 38]. For the case $\mu_k > 0$, regularized limited-memory schemes may in principle be employed [27]. Alternatively, one may adopt the simple and robust approximation mentioned in the same reference. Specifically, we approximate B_k in eq. (3.39) by

$$B_k \approx \mathrm{BFGS}(\{s_{k_i}, \bar{y}_{k_i} + \mu_k s_{k_i}\}_{i=0}^{p-1}) - \mu_k I, \tag{3.41}$$

so that $B_k + \mu_k I \approx \mathrm{BFGS}(\{s_{k_i}, \bar{y}_{k_i} + \mu_k s_{k_i}\}_{i=0}^{p-1})$. Its inverse can then be computed using the standard two-loop recursion. Although this approximation is inexact, it has been reported to yield nearly identical practical performance while remaining comparatively simple to implement [27]. We adopt this approximation in our numerical experiments.

Separately, there exist function-based modified secant condition [2, 24, 54, 58] that adjusts the secant equation using available function values when those quantities are reliable. Such adjustments aim to improve practical convergence behavior. We partially adopt these function-based modifications following prior work [55, 59]. The suffix `-MS` in section 3.5 indicates the use of this technique. Because these modifications are well established, we omit low-level implementation details here. Further specifics are available in the cited references and in our open-source implementation[*1].

3.4.2 Step Size α_k

We next discuss the adjustment of the step size α_k in line 4 of Algorithm 2.

As indicated in the pseudo-code, we choose α_k using interpolation of the objective function. Moré–Thuente line searches [35] are widely used in quasi-Newton methods [49], and they typically employ cubic or quadratic interpolation to select trial step sizes. In our implementation, we use an essentially identical procedure with a slight modification, namely, clipping the trial step size to the interval $[\beta_{\min}\alpha_k, \beta_{\max}\alpha_k]$. For all numerical experiments, we set the Armijo parameter to $c = 10^{-4}$ and choose the interpolation bounds as $\beta_{\min} = 1/16$ and $\beta_{\max} = 15/16$.

Furthermore, when $\mu_k > 0$ and $\alpha_k = 1$, we introduce a one-time adjustment of the trial step size using gradient information. Let d_k be the search direction and g_k and g_{try} the gradients at the current and trial points. If the directional derivative along d_k changes sign, $d_k^\top g_k < 0$ and $d_k^\top g_{\text{try}} > 0$, and the trial gradient is sufficiently aligned with d_k, namely $d_k^\top g_{\text{try}} > 0.5\|d_k\|\|g_{\text{try}}\|$, we rescale α_k once by a secant-like factor,

$$\alpha_k \leftarrow \operatorname{clip}\left(\alpha_k \frac{-d_k^\top g_k}{d_k^\top g_{\text{try}} - d_k^\top g_k}, \beta_{\min}\alpha_k, \beta_{\max}\alpha_k\right),$$

where $\operatorname{clip}(x,a,b) := \min(\max(x,a),b)$ is a clipping function. When objective values are unreliable and backtracking and interpolation do not occur, this correction effectively refines the step size. Since this adjustment is at most once per iteration, the theoretical results in section 3.3 remain valid.

3.4.3 Regularization Parameter μ_k

We finally discuss the adjustment of the regularization parameter μ_k in line 8 of Algorithm 1.

In our implementation, we choose the regularization parameter as

$$\mu_k = \operatorname{clip}\left(\frac{1}{10}\|g_k\|, \frac{1}{100}G_k, G_k\right) \quad \text{where} \quad G_k := \sqrt{\varsigma + \sum_{j\in K, j\le k} \|g_j\|^2},$$

which is consistent with the general form (eq. (3.6)) with an adaptive factor $\theta_k \in [10^{-2}, 1]$. In practice, we may set $\varsigma = 0$ as long as the initial gradient is nonzero since ς is mainly introduced for avoiding zero division, but to make it consistent with the theory, we set $\varsigma = 10^{-10}$ in all experiments.

When B_k is a good approximation of the Hessian, the regularization parameter μ_k should be small for fast convergence. Since the quantity G_k is strictly increasing with k, it is natural in practice to decrease the effective regularization when sufficient progress in the objective function is achieved. When the difference between the two sides of eq. (3.5) is larger than 1, we restart the accumulation by resetting K^+ to the empty set and G_k to ς. As long as the number of such restarts is finite, the overall convergence analysis in section 3.3 remains valid. Since these choices are heuristic, there may be room for further improvement. Especially, further increasing μ_k stabilizes the method when the objective function values are highly noisy, and decreasing μ_k more aggressively may accelerate convergence when the objective function is accurate.

3.5 Numerical Experiments

3.5.1 Methods

We compared the following optimization algorithms:

- (`Ours`): Our proposed regularized quasi-Newton method
  - We used Algorithms 1 and 2 with the details in section 3.4.
  - (`Ours-MS`) means the variant incorporating the modified secant condition described in section 3.4.1.
- (`Line`): Line search L-BFGS method
  - Our Python implementation ported from LBFGSpp [39].
  - The step size was determined using the Moré–Thuente line search, closely resembling SciPy's implementation.
  - (`Line-MS`) means the same variant as `Ours-MS`.

- (`Reg`): Regularized L-BFGS method [27]
  - A quadratic regularization-based quasi-Newton method that does not rely on line search and is conceptually closer to trust-region methods.
  - We wrapped the function `regLBFGS` with `solveNonmonotone` from [27].
  - (`Reg-Sec`) means the incorporation of the approximated computation as in eq. (3.41). This is a wrapped function of `regLBFGSsec` from [27].
- (`SciPy`): SciPy's L-BFGS-B method [49]
  - An established L-BFGS method implemented in C and called from Python.
  - We used the default parameters, with the exception that `ftol` was set to 0 to avoid premature termination.
- (`NTQN`): Noise-tolerant quasi-Newton method [45, 53]
  - A method designed to accommodate inexact gradient information, as mentioned in section 3.1.3.
  - We mainly used the default parameter settings, and slightly modified the stopping criteria to terminate at the desired gradient norm tolerance.

In the following, each method is referred to by the name given in parentheses. All these methods are L-BFGS-based, and we set the number of stored curvature pairs to 10, the default value in SciPy's L-BFGS-B implementation.

All experiments were conducted on a Microsoft Windows 11 Home system (version 10.0, build 26100) equipped with a 13th Gen Intel® Core™ i7-1360P CPU, featuring 12 physical cores and 16 logical processors. All computations were performed using the CPU only.

3.5.2 Experimental Results

Test Problems and Settings

We evaluated all algorithms on unconstrained problems from the CUTEst benchmark collection [18], implemented in Python using the PyCUTEst interface [16]. All problems were initialized with the default starting points provided by the library. Following the previous work [27], we excluded problems such as the initial point being already stationary or containing `NaN` and `INF` in its function values or gradient. To simulate different numerical environments, we tested the following four settings.

- Artificially noised 64-bit floating-point precision (220 problems, $\varepsilon_{\mathrm{f}} = 10^{-2}$)
- 64-bit floating-point precision (220 problems, $\varepsilon_{\mathrm{f}} = 2.22 \times 10^{-9}$)
- 32-bit floating-point precision (220 problems, $\varepsilon_{\mathrm{f}} = 1.19 \times 10^{-2}$)
- 16-bit floating-point precision (152 problems, $\varepsilon_{\mathrm{f}} = 9.77 \times 10^{-1}$)

In the artificial noise setting, we added uniform random noise to both the objective function values and each component of the gradient vectors independently, i.e.,

$$\overline{f}(x) = f(x) + \mathrm{unif}(-10^{-3}, 10^{-3}), \quad \nabla\overline{f}(x) = \nabla f(x) + \mathrm{unif}(-10^{-3}, 10^{-3})\mathbb{1},$$

where $\mathrm{unif}(a,b)$ denotes a uniform random variable in the interval $[a,b]$ and $\mathbb{1}$ is the vector of all ones. This setting is the same as the one in [45]. In the reduced precision settings, to simulate lower numerical precision, we first cast the input vector x to the target lower precision, then computed $\overline{f}(x)$ and $\nabla\overline{f}(x)$ using the lower-precision input. Although the internal computations of CUTEst remain in 64-bit precision, the casting of the input vector effectively simulated the impact of reduced precision on function and gradient evaluations.

Each setting uses a different error rate ε_{f} in the error model (eq. (3.1)), as summarized in Table 3.1. The table relates the machine epsilon (maximum relative rounding error) for each floating-point precision to the ε_{f} value used in our experiments. The relatively large ε_{f} values are chosen to account for the cumulative errors that may arise during optimization. We also set the

explanation	min abs value	relative error	ε_{f}
64-bit double precision	2.23×10^{-308}	$2^{-52} \approx 2.22 \times 10^{-16}$	2.22×10^{-9}
32-bit single precision	1.18×10^{-38}	$2^{-23} \approx 1.19 \times 10^{-7}$	1.19×10^{-3}
16-bit half precision	6.10×10^{-5}	$2^{-10} \approx 9.77 \times 10^{-4}$	9.77×10^{-2}

Table 3.1: The "min abs value" is the smallest positive normalized number, the "relative error" is the maximum relative rounding error, and ε_{f} is the parameter in (**??**) used in our experiments.

maximum number of iterations to $k_{\max} = 15{,}000$ and timed out runs exceeding 10 minutes per problem.

Performance Profiles

We used performance profiles [14] to compare the algorithms. Let P denote the set of test problems and S the set of solvers. For each problem $p \in P$ and solver $s \in S$, let $n_{p,s} > 0$ denote the number of function and gradient evaluations required to reach a specified gradient norm tolerance $\varepsilon_{\mathrm{gtol}}$. Following common implementations [49], we used the infinity norm for the gradient, though other norms may equally be employed. Note that $n_{p,s}$ is not time in this experiment but the number of evaluations. When a solver fails to converge within the gradient tolerance or exceeds a maximum number of evaluations, we set $n_{p,s} = +\infty$. Then, the performance ratio $r_{p,s}$ is defined as

$$r_{p,s} = \frac{n_{p,s}}{\min_{s' \in S} n_{p,s'}}.$$

For a given threshold $\tau \geq 1$ and solver s, the performance profile plots the proportion of problems for which $r_{p,s} \leq \tau$. This means that the y-coordinate for solver s at x-coordinate τ is given by

$$\rho_s(\tau) = \frac{1}{|P|} \left| \{ p \in P \mid r_{p,s} \leq \tau \} \right|.$$

A curve that lies above others indicates better performance.

Results with Artificial Noise

We first present the result in the artificial noise setting described in section 3.5.2. Figure 3.1 presents the performance profile. In this regime, the proposed method shows superior robustness compared to other quasi-Newton methods. This result highlights the validity of our theoretical analysis in section 3.3 and demonstrates the effectiveness of our approach in handling noisy objective function values.

Results at Different Precision Levels

We next present performance profiles for each floating-point precision (64-bit, 32-bit, and 16-bit). We evaluated each method at three gradient norm tolerance levels, i.e., $\varepsilon_{\mathrm{gtol}} \in \{10^{-1}, 10^{-3}, 10^{-5}\}$. These tolerance levels represent different optimization regimes, from moderate accuracy to high precision requirements. Figure 3.2 shows the comprehensive performance profiles across all three precision levels. Our methods (`Ours`) and (`Ours-MS`) demonstrate competitive or superior performance compared to existing methods in the standard 64-bit setting and maintain robust performance as precision decreases to 32-bit and 16-bit. The proposed method maintains stability even when traditional line search methods encounter difficulties with reduced precision arithmetic.

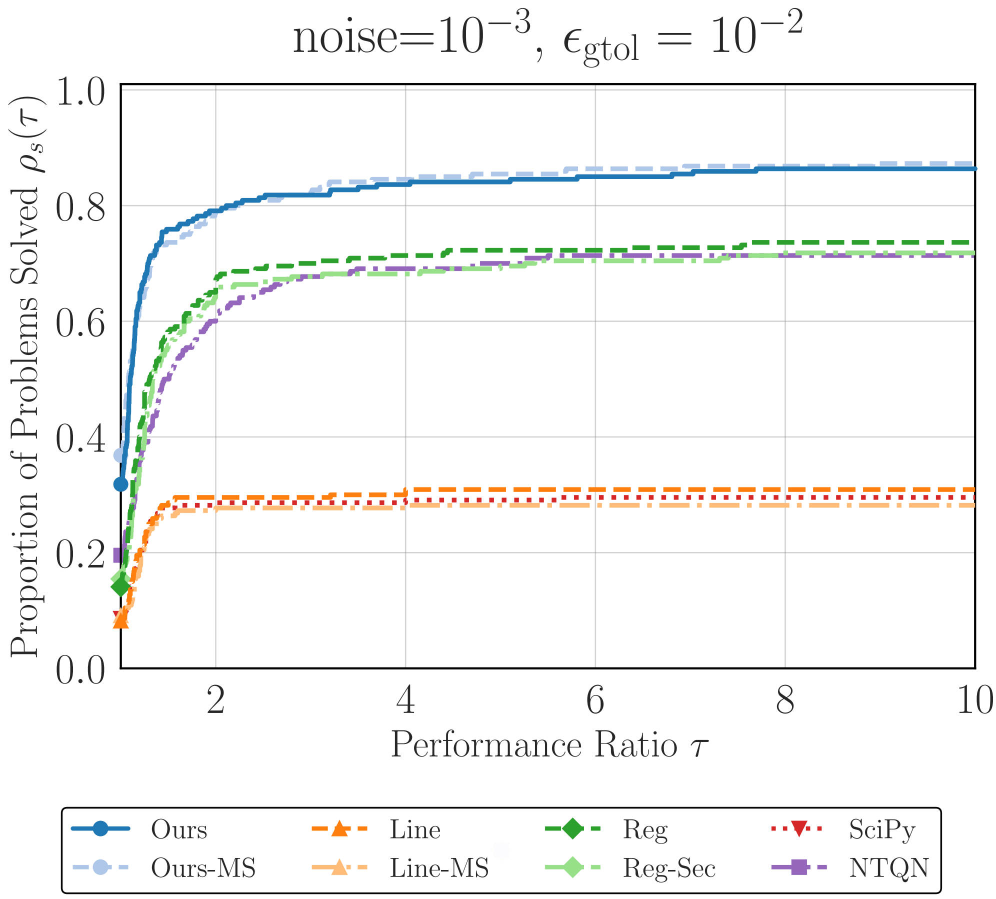


Fig. 3.1: Performance profile for artificial noise setting (level of 10^{-3}) and $\varepsilon_{\text{gtol}} = 10^{-2}$.

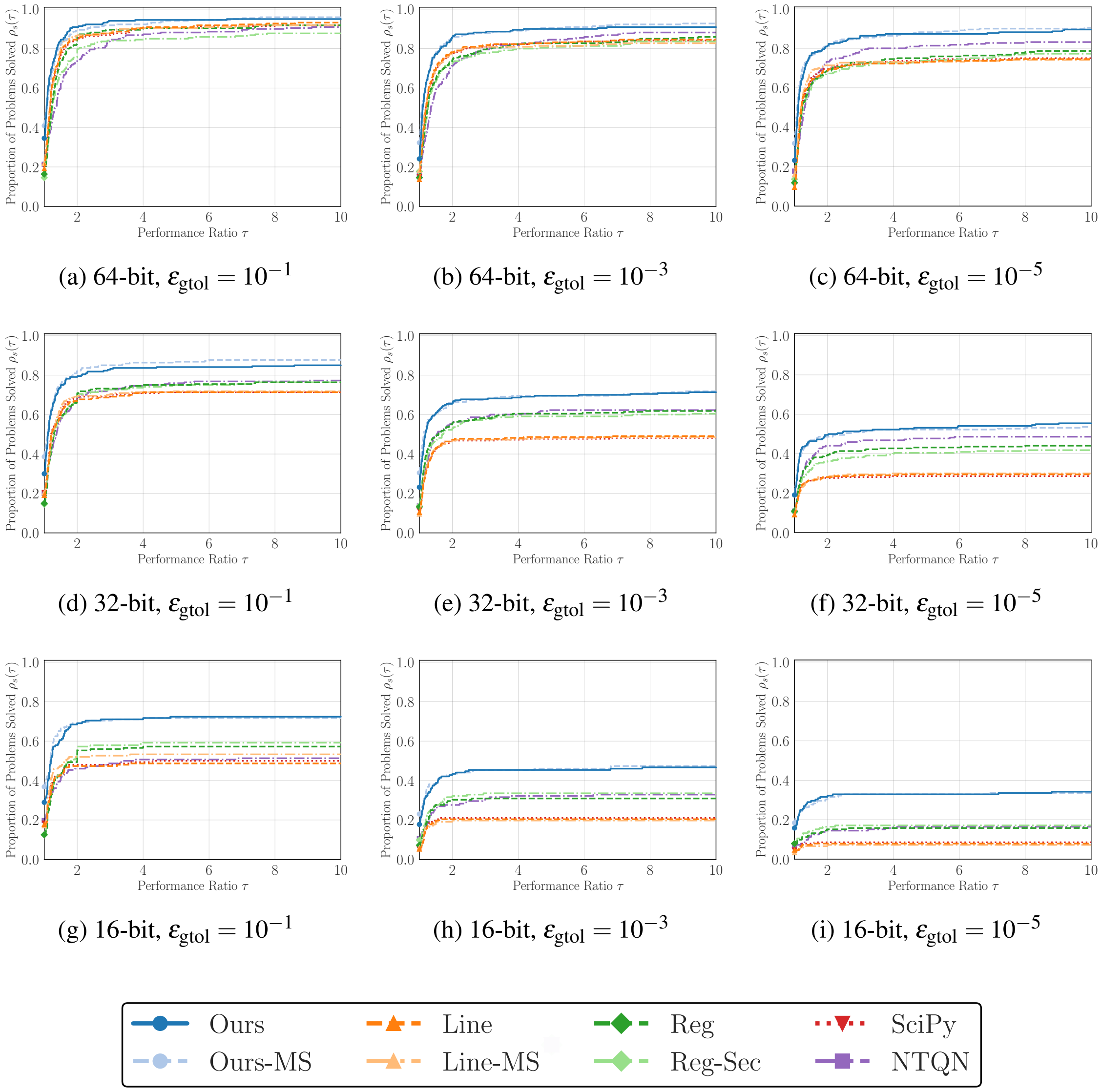
3.5.3 Computational Time per Iteration

In this subsection, we present an experiment to compare the per-iteration computational cost of each method. Here, our purpose lies in the measurement of the total execution time, not the number of oracle calls. We conducted experiments on the ill-conditioned quadratic problem

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2} \sum_{i=1}^{n} i x_i^2$$

with $n = 10{,}000$. The initial point x_0 was chosen as the all-ones vector. We used memory size $m = 10$ and maximum iterations $k_{\max} = 100$, and we did not set any stopping criteria other than reaching $k_{\max}$. The simplicity of the optimization problem and the small value of $k_{\max}$ are well suited to the purpose of this experiment, as they ensure that the total runtime is dominated by algorithmic overhead rather than function or gradient computation. Timing data were collected after three warm-up runs, followed by 100 recorded trials.

In Figure 3.3, we report the execution time statistics, and we can observe that our proposed methods (`Ours`) and (`Ours-MS`) exhibit competitive performance compared to other methods. Combining with the results from section 3.5.2, our methods are competitive or superior in total oracle calls and per-iteration computational cost, indicating practical efficiency. As a side note, although the (`SciPy`) calls optimized C routines, it incurs associated dispatch overhead, produc-



(a) 64-bit, $\varepsilon_{\text{gtol}} = 10^{-1}$ (b) 64-bit, $\varepsilon_{\text{gtol}} = 10^{-3}$ (c) 64-bit, $\varepsilon_{\text{gtol}} = 10^{-5}$

(d) 32-bit, $\varepsilon_{\text{gtol}} = 10^{-1}$ (e) 32-bit, $\varepsilon_{\text{gtol}} = 10^{-3}$ (f) 32-bit, $\varepsilon_{\text{gtol}} = 10^{-5}$

(g) 16-bit, $\varepsilon_{\text{gtol}} = 10^{-1}$ (h) 16-bit, $\varepsilon_{\text{gtol}} = 10^{-3}$ (i) 16-bit, $\varepsilon_{\text{gtol}} = 10^{-5}$

Fig. 3.2: Performance profiles across different floating-point precisions (64, 32, and 16-bit) and tolerance levels. The results demonstrate the robustness of the proposed methods.

ing a larger mean time. One of the reasons why the (`Reg`) and (`Reg-Sec`) methods take longer execution time is due to the implementation aspect of the original code.

In Table 3.2, we report the total number of oracle calls (calls of $\overline{f}$ and $\nabla\overline{f}$) and final objective values $f(x_{k_{\max}})$. The purpose of this table is to confirm that all methods achieved similar convergence behavior within the limited number of iterations. The comparable values in the table indicate that the comparison of execution times is fair and not affected by differences in convergence behavior.

3.6 Conclusion

In this work, we proposed a regularized quasi-Newton method for noisy nonconvex optimization and established its global convergence properties. We also demonstrated the method's effectiveness through numerical experiments on standard benchmark problems under various noise levels

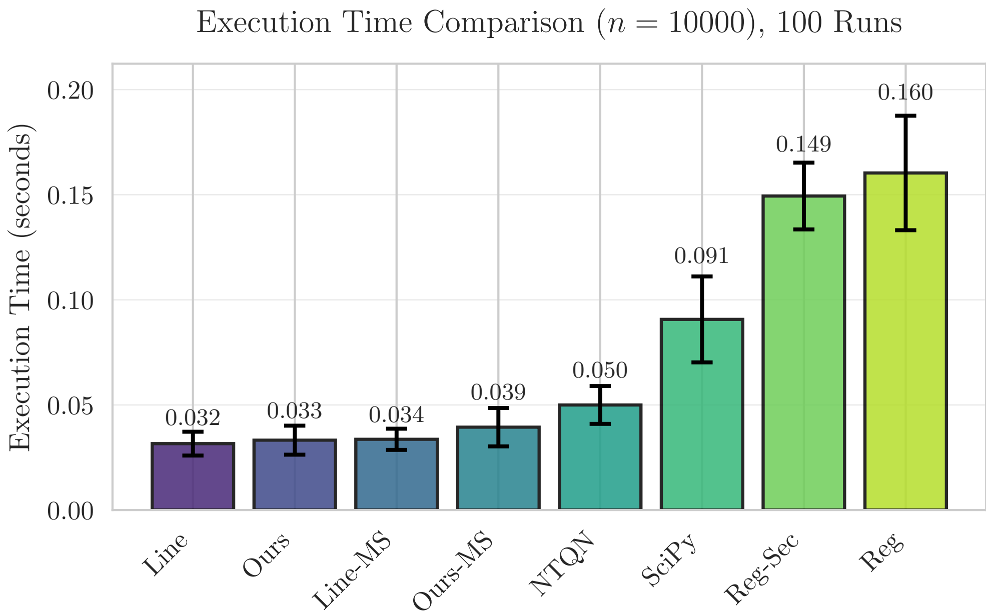


Fig. 3.3: Mean execution times for the experiment in section 3.5.3. This plot roughly indicates the extra computational overhead of each method. Error bars represent one standard deviation over 100 runs.

	`Line`	`Ours`	`Line-MS`	`Ours-MS`	`NTQN`	`SciPy`	`Reg-Sec`	`Reg`
# Calls	212	202	212	202	219	212	206	206
$\overline{f}(x_{k_{\max}})$	1.24	1.34	1.24	1.34	1.43	1.24	1.48	1.47

Table 3.2: The number of oracle calls and final observed objective values for the experiment in section 3.5.3.

and precision settings. The results indicate that our methods are robust and effective, particularly in noisy or low-precision environments where traditional line search methods may struggle.

Future work includes the investigation of local convergence properties, the extension to constrained optimization, and the applications to practical problems in machine learning and scientific computing. Additionally, exploring adaptive strategies for selecting algorithmic parameters based on problem characteristics could further enhance the method's performance. An explicit analysis of the effect of inexact gradient evaluations would also provide valuable insight into the method's robustness and guide further enhancements.

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A

Appendix

A.1 Proof of Lemma 6

The proof of Lemma 6 is similar to existing results [51, Lemma 3.2], [23, Lemma 3.1].

Proof. Recall that eq. (3.6) defines μ_k as $\theta_k\sqrt{\varsigma+\sum_{j\in K^+,j\le k}\|g_j\|^2}$, and line 8 of Algorithm 1 asserts $\theta_k\in[\theta_{\min},\theta_{\max}]$. Let $k_i\in K^+$ be the i-th element of K^+. For any $0<x<y$, $x/y\le -\ln(1-x/y)=\ln y-\ln(y-x)$ holds. Thus, we have

$$\begin{aligned}\frac{\|g_{k_i}\|^2}{\mu_{k_i}^2} &\le \frac{1}{\theta_{\min}^2}\frac{\|g_{k_i}\|^2}{\varsigma+\sum_{\ell=0}^{i}\|g_{k_\ell}\|^2} && \text{(definition)}\\ &\le \frac{1}{\theta_{\min}^2}\left(\ln\left(\varsigma+\sum_{\ell=0}^{i}\|g_{k_\ell}\|^2\right)-\ln\left(\varsigma+\sum_{\ell=0}^{i-1}\|g_{k_\ell}\|^2\right)\right). && (x/y\le\ln y-\ln(y-x))\end{aligned}$$

Similarly, $x/\sqrt{y}\ge x/(\sqrt{y}+\sqrt{y-x})=\sqrt{y}-\sqrt{y-x}$ holds. Thus, we have

$$\begin{aligned}\frac{\|g_{k_i}\|^2}{\mu_{k_i}} &\ge \frac{1}{\theta_{\max}}\frac{\|g_{k_i}\|^2}{\sqrt{\varsigma+\sum_{\ell=0}^{i}\left\|g_{k_\ell}^2\right\|}} && \text{(definition)}\\ &\ge \frac{1}{\theta_{\max}}\left(\sqrt{\varsigma+\sum_{\ell=0}^{i}\|g_{k_\ell}\|^2}-\sqrt{\varsigma+\sum_{\ell=0}^{i-1}\|g_{k_\ell}\|^2}\right). && (x/\sqrt{y}\ge\sqrt{y}-\sqrt{y-x})\end{aligned}$$

Summing these inequalities over $0\le i<|K^+|$ yields the claim. □

A.2 Proof of Proposition 2

The proof of Proposition 2 is similar to existing results [17, Theorem 5.5] [19, Lemma 1].

Proof. We consider standard BFGS updates with a single curvature pair (s,y). From eqs. (3.36) and (3.37), we have

$$\left\|\frac{yy^\top}{y^\top s}\right\|=\frac{y^\top y}{y^\top s}\le\Lambda,\quad \left\|\frac{ys^\top}{y^\top s}\right\|\le\frac{\|y\|\|s\|}{\sqrt{\frac{1}{\Lambda}\|y\|^2}\sqrt{\lambda\|s\|^2}}=\sqrt{\kappa},\quad \left\|\frac{ss^\top}{y^\top s}\right\|=\frac{s^\top s}{y^\top s}\le\frac{1}{\lambda}. \tag{A.1}$$

Let B, H denote the positive definite Hessian approximation before the update and its inverse, and B_+, H_+ the updated matrices. The BFGS update rule is

$$B_+ = B - \frac{Bss^\top B}{s^\top Bs} + \frac{yy^\top}{y^\top s}, \quad H_+ = \left(I - \frac{sy^\top}{y^\top s}\right) H \left(I - \frac{ys^\top}{y^\top s}\right) + \frac{ss^\top}{y^\top s}.$$

Since B is symmetric positive definite matrix and $s \neq 0$, we have

$$\|B_+\| \leq \|B\| + \left\|\frac{yy^\top}{y^\top s}\right\| \leq \|B\| + \Lambda, \tag{A.2}$$

$$\begin{aligned}
\|H_+\| &= \left\|\left(I - \frac{ys^\top}{y^\top s}\right)^\top H \left(I - \frac{ys^\top}{y^\top s}\right) + \frac{ss^\top}{y^\top s}\right\| \\
&\leq \left\|I - \frac{ys^\top}{y^\top s}\right\|^2 \|H\| + \left\|\frac{ss^\top}{y^\top s}\right\| \\
&\leq \left(1 + \left\|\frac{ys^\top}{y^\top s}\right\|\right)^2 \|H\| + \left\|\frac{ss^\top}{y^\top s}\right\| \\
&\leq \left(1 + \sqrt{\kappa}\right)^2 \|H\| + \frac{1}{\lambda}. \qquad \text{(eq. (A.1))}
\end{aligned} \tag{A.3}$$

Applying eqs. (A.2) and (A.3) recursively over p updates and using the initial matrix γI, we obtain

$$\begin{aligned}
\left\|\mathrm{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1})\right\| &\leq \gamma + p\Lambda, \\
\left\|(\mathrm{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1}))^{-1}\right\| &\leq (1+\sqrt{\kappa})^{2p}\gamma + \frac{1}{\lambda}\sum_{i=0}^{p-1}(1+\sqrt{\kappa})^{2i} \\
&= (1+\sqrt{\kappa})^{2p}\gamma + \frac{1}{\lambda}\frac{(1+\sqrt{\kappa})^{2p}-1}{(1+\sqrt{\kappa})^2-1} \\
&\leq (1+\sqrt{\kappa})^{2p}\left(\gamma + \frac{1}{\lambda\left(2\sqrt{\kappa}+\kappa\right)}\right).
\end{aligned}$$

Therefore, $\mathrm{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1})$ satisfies the stated bounds in eq. (3.38). □

The conditions eqs. (3.36) and (3.37) are standard in the analysis of quasi-Newton methods [8, Theorem 2.1]. If f is L-smooth (Assumption 2) and convex, then by the Baillon–Haddad theorem [3, 4] its gradient is $1/L$-cocoercive:

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq \frac{1}{L}\|\nabla f(x) - \nabla f(y)\|^2 \tag{A.4}$$

for all $x, y \in \mathbb{R}^n$. Applying this inequality with $x = x_{k+1}$ and $y = x_k$ yields eq. (3.36) with $\Lambda = L$.